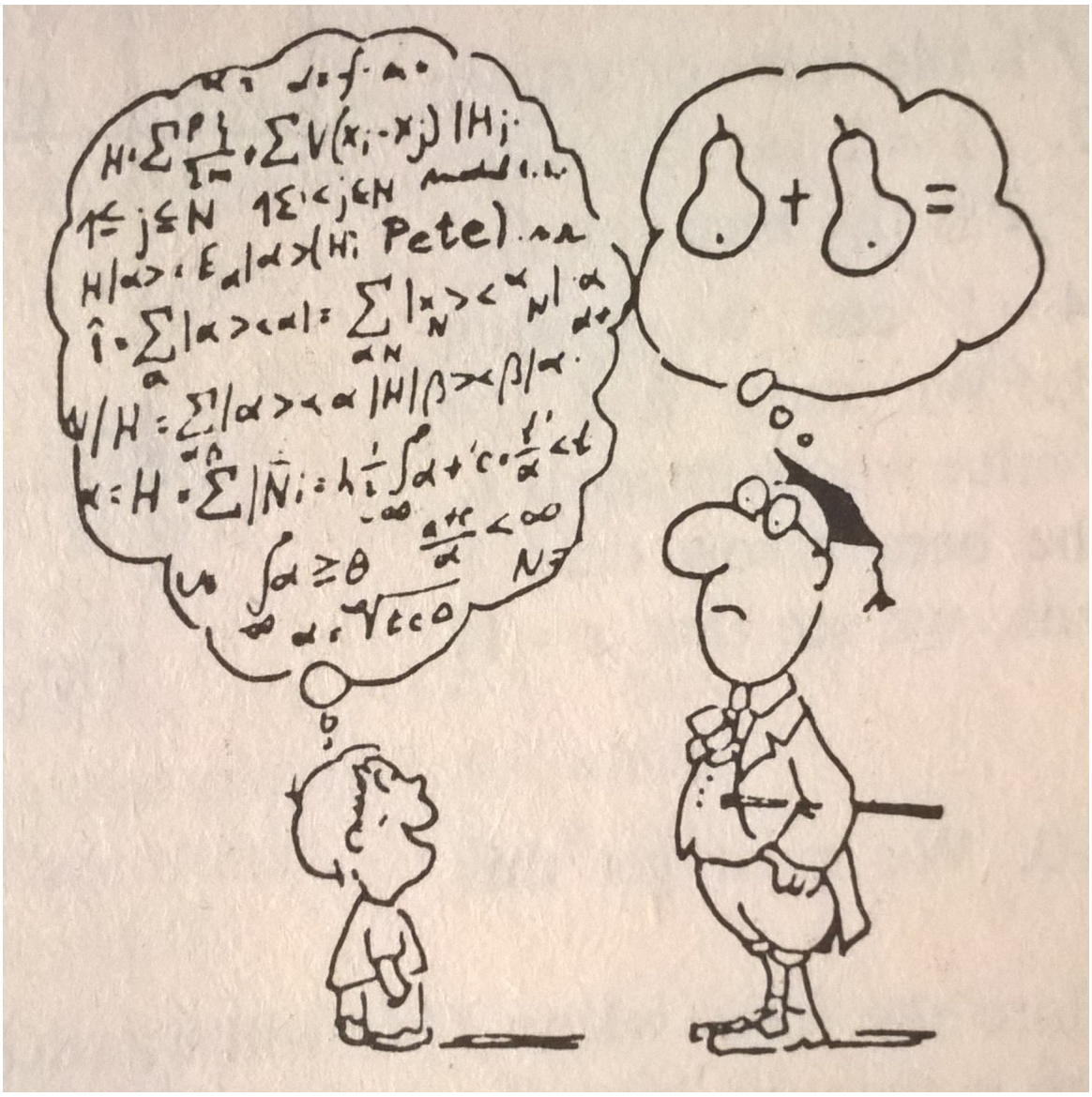
MATH BASICS


Pete

1 Definition

1.1 Exponents

1.

$$a^n \;=\; a \times a \times a \times a \;....\; n \quad factors \qquad (n \in \mathbb{N},\; a \in \mathbb{R})$$

2.

$$a^{-m} \;=\; \frac{1}{a^m} \qquad (m \in \mathbb{Z}^+,\; a \in \mathbb{R},\; a \neq 0)$$

$$and:$$

$$\frac{1}{a^{-m}} \;=\; a^m$$

3.

$$a^0 = 1 \qquad (a \in \mathbb{R},\; a \neq 0)$$

1.2 Rational Exponents:

1.

$$\sqrt[n]{a} \;=\; r \qquad (a \;>\; 0,\; n \in \mathbb{N},\; n \geqslant 2,\; r \;>\; 0),\; \Longleftrightarrow\; r^n \;=\; a$$

2.

$$a^{\frac{1}{n}} \;=\; \sqrt[n]{a}; \qquad (a \;>\; 0,\; n \;\geqslant 2,\; n \in \mathbb{N})$$

3.

$$a^{\frac{-1}{n}} \;=\; \sqrt[n]{a^{-1}}; \qquad (a \;>\; 0,\; n \;> 0,\; n \in \mathbb{N})$$

4.

$$a^{\frac{m}{n}} \;=\; \sqrt[n]{a^m}; \qquad (a \;>\; 0;\; m,\; n \in \mathbb{Z},\; n \geqslant 2)$$

2 Law

2.1 Exponents

1.
$$a^m \times a^n = a^{m+n} \qquad (m,\ n \in \mathbb{N})$$
$$a^m \times a^n = a^{m+n} \qquad (m,\ n \in \mathbb{Z};\ a \neq 0,\ if \quad m \quad or \quad n < 0)$$

2.
$$\frac{a^m}{a^n} = a^{m-n} \qquad (m,\ n \in \mathbb{Z};\ a \in \mathbb{R};\ a \neq 0)$$

3.
$$(ab)^m = a^m\, b^m \qquad (m \in \mathbb{Z})$$

4.
$$(a^m)^n = a^{mn} \qquad (m,\ n \in \mathbb{Z})$$

2.2 Rational Exponents

1.
$$a^r \times a^t = a^{r+t} \qquad (a > 0; \quad r,\ t \in \mathbb{Q})$$

2.
$$\frac{a^r}{a^t} = a^{r-t} \qquad (a > 0; \quad r,\ t \in \mathbb{Q})$$

3.
$$(a^t)^r = a^{tr} \qquad (a > 0, \quad t,\ r \in \mathbb{Q})$$

4.
$$(ab)^t = a^t\, b^t\ ; \quad \left(\frac{a}{b}\right)^t = \frac{a^t}{b^t}\ ; \qquad (a,\ b > 0, \quad t \in \mathbb{Q})$$
$$and:$$
$$a^t\, b^t = (ab)^t \qquad and \qquad \frac{a^t}{b^t} = \left(\frac{a}{b}\right)^t$$

2.3 Distributive law

$$a(b + c) = ab + ac$$

$$\begin{aligned}(a + b)(c + d) &= (a + b)c + (a + b)d \\ &= ac + bc + ad + bd\end{aligned}$$

$$A^2 - B^2 = (A - B)(A + B)$$

2.4 Commutative law

$$ab = ba$$

3 Properties

3.1 Addition

$$0 + a = a$$

$$\frac{a}{b} + \frac{c}{b} = \frac{a + c}{b} \qquad (b \neq 0)$$

$$\frac{a}{b} - \frac{c}{b} = \frac{a - c}{b} \qquad (b \neq 0)$$

3.2 Multiplication

$$0 \times a = 0$$

$$\frac{0}{a} = 0 \times \frac{1}{a} = 0$$

$$1 \times a = a$$

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd} \qquad (b \neq 0;\ d \neq 0)$$

3.3 Division

$$\frac{a}{0} = undefined$$

$$\frac{p}{q} \div \frac{r}{s} = \frac{p}{q} \times \frac{s}{r} = \frac{ps}{qr} \qquad (q \neq 0;\ r \neq 0;\ s \neq 0)$$

4 Methods

HCF - highest common factor (pôr variável em evidência)

Factorisation

LCD or LCM - Lowest common denominator or lowest common multiple

Capítulo 1

Definições

Definição 1 *Capacitância*

$$
\begin{aligned}
Q_c(t) =& \int^t i(t) \quad dt \\
=& Q_c(0^-) + \int_{0^-}^t i(t) \quad dt \\
V_c(t) =& \frac{Q_c(t)}{C} \\
=& \frac{1}{C} \quad \int^t i_c(t) \quad dt \\
=& \frac{Q_c(0^-)}{C} + \frac{1}{C} \quad \int_0^t i_c(t) \quad dt \\
=& V(0^-) + \frac{1}{C} \quad \int_0^t i_c(t) \quad dt \\
i_c(t) =& C \quad \frac{dV_c(t)}{dt} \\
W =& \frac{1}{2}\, C\, V^2
\end{aligned}
$$

Definição 2 *Indutância*

$$
\begin{aligned}
\psi_L(t) =& \int^t V_L(t) \quad dt \\
=& \psi_L(0^-) + \int_{0^-}^t V_L(t) \quad dt \\
V_L(t) =& L \quad \frac{di_L(t)}{dt} \\
i_L(t) =& \frac{\psi_L(t)}{L} \\
=& \frac{1}{L} \quad \int^t V_L(t) \quad dt \\
=& \frac{\psi_L(0^-)}{L} + \frac{1}{L} \quad \int_0^t V_L(t) \quad dt \\
=& i_L(0^-) + \frac{1}{L} \quad \int_0^t V_L(t) \quad dt \\
W =& \frac{1}{2}\, L\, i^2
\end{aligned}
$$

Definição 3 *Resistência*

$$
\begin{aligned}
V_R(t) &= R \quad i_R(t) \\
i_R(t) &= \frac{V_R(t)}{R} \\
P &= Ri^2 \\
P &= \frac{U^2}{R} \\
W &= P\,\Delta t
\end{aligned}
$$

Definição 4 *Valor Médio*

$$X_{av} = \frac{1}{T}\int_0^T X(t)dt$$

Definição 5 *Valor Eficaz*

$$X_{ef} = \sqrt{\frac{1}{T}\int_0^T X^2(t)dt}$$

Eng. Sérgio Santos

Anexo A

Definições

Definição 6 *Capacitância*

$$
\begin{aligned}
Q_c(t) =& \int^t i(t) \quad dt \\
=& Q_c(0^-) + \int_{0^-}^t i(t) \quad dt \\
V_c(t) =& \frac{Q_c(t)}{C} \\
=& \frac{1}{C} \quad \int^t i_c(t) \quad dt \\
=& \frac{Q_c(0^-)}{C} + \frac{1}{C} \quad \int_0^t i_c(t) \quad dt \\
=& V(0^-) + \frac{1}{C} \quad \int_0^t i_c(t) \quad dt \\
i_c(t) =& C \quad \frac{dV_c(t)}{dt} \\
W =& \frac{1}{2}\, C\, V^2
\end{aligned}
$$

Definição 7 *Indutância*

$$
\begin{aligned}
\psi_L(t) =& \int^t V_L(t) \quad dt \\
=& \psi_L(0^-) + \int_{0^-}^t V_L(t) \quad dt \\
V_L(t) =& L \quad \frac{di_L(t)}{dt} \\
i_L(t) =& \frac{\psi_L(t)}{L} \\
=& \frac{1}{L} \quad \int^t V_L(t) \quad dt \\
=& \frac{\psi_L(0^-)}{L} + \frac{1}{L} \quad \int_0^t V_L(t) \quad dt \\
=& i_L(0^-) + \frac{1}{L} \quad \int_0^t V_L(t) \quad dt \\
W =& \frac{1}{2}\, L\, i^2
\end{aligned}
$$

Definição 8 *Resistência*

$$
\begin{aligned}
V_R(t) &= R \quad i_R(t) \\
i_R(t) &= \frac{V_R(t)}{R} \\
P &= Ri^2 \\
P &= \frac{U^2}{R} \\
W &= P\,\Delta t
\end{aligned}
$$

Definição 9 *Valor Médio*

$$X_{av} = \frac{1}{T}\int_0^T X(t)dt$$

Definição 10 *Valor Eficaz*

$$X_{ef} = \sqrt{\frac{1}{T}\int_0^T X^2(t)dt}$$

Anexo B

Física

Força [N] [Kgf]

$$\sum F_{(t)} = M\ a_{(t)} = M\ \ddot{x}_{(t)}$$

$$\sum F_R = \sum F_{action} - \sum F_{reaction}$$

$$f_{(t)} = -K\ x_{(t)}$$
$$f_{(t)} = -B\ \dot{x}_{(t)}$$

Torque [N.m]

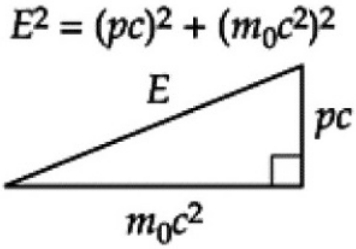
$$\sum T_{(t)} = J\ \gamma_{(t)} = M\ \ddot{\theta}_{(t)}$$

$$\sum T_R = \sum T_{action} - \sum T_{reaction}$$

$$T_{(t)} = -K\ \theta_{(t)}$$
$$T_{(t)} = -B\ \dot{\theta}_{(t)}$$
$$T = F \times r$$

Energia [Joule]

$$W = F\ d$$
$$W = P\ \Delta t$$
$$E = M\ C^2$$



[?]

Energia Cinética [Joule]

$$E_c = \frac{1}{2}\ m\ v^2$$

Energia Potencial [Joule]

$$E_p = m\ g\ h$$

Energia Térmica

$Q - Heat \quad energy$
$Q_{(t)} - temperature$
$R - heat \quad resistance$

$$Q = \frac{Q_{1(t)} - Q_{2(t)}}{R}$$